Extend/Retract Screed Extension

The screed operator can extend or retract screed extensions on either or both sides of the paver using the corresponding extend or retract buttons on the screed control box. Arrows on each symbol point the direction the screed mechanism moves.



- Press and hold the corresponding button to extend or retract the screed to desired width.
- Release button when desired width is achieved.

Adjust the screed while paving using the following procedures:

1. Move paver to the starting position for paving.
2. Extend the screed extensions to the desired width.
3. Level the screed using the tow points. ***(Page 4-59)***
   - Refer to the elevation gauge. Raise or lower until the rod end of the cable is flush with “0” on the gauge.
   - While paving, refer to both gauges and make minor adjustments if needed.
4. Null the screed on starting surface (or starter blocks) using flight screws. Make adjustments as needed while paving.
5. On the first pass, adjust the end gate so the front tilts down slightly when the screed is lifted. This will allow the end gate to set itself to the grade.
   - If the end gate digs into the asphalt at the front, adjust it until the end gate tilts back more.
6. On the first pass, leave approximately six to eight inches of unrolled asphalt where the joint will be made.
7. When making a joint, the end gate must be set to where it fits flush with the bottom of the screed.
   - If the joint looks too high or too low, adjust the flight screw on the screed one turn at a time and allow four to five feet of travel to correct itself.

NOTE: Too much adjustment up or down may cause uneven pavement.

NOTE: Keep runners clean. When making a joint, spray solvent on the runners.

Horizontal Depth Screws

The screed operator can also use the horizontal depth screws for making minor adjustments to mat thickness for the main screed as needed while paving. The depth screws also have a locking lever mechanism to lock the screw in place as needed. **(Figure 4-67)**

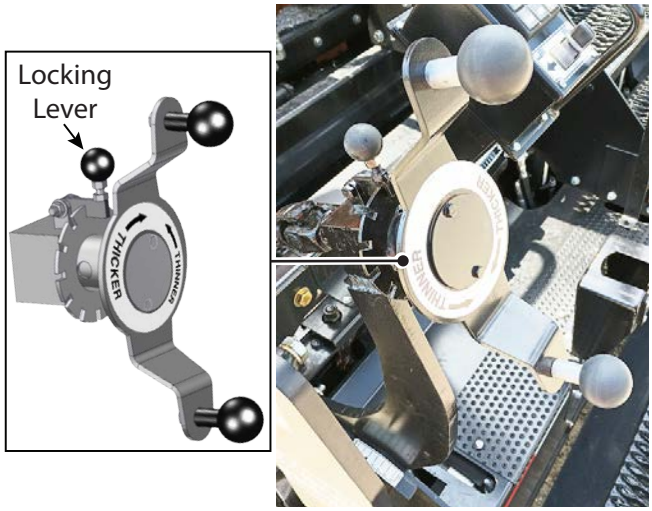


Figure 4-67. Horizontal Depth Screw

You must be operating tow point cylinders in the manual mode when using the depth screw

1. Before paving, refer to the screed elevation gauge. **(Figure 4-68)**
2. Raise or lower the screed until the cable rod end is flush with “0”on the indicator decal.
3. Slowly move both joysticks forward.
4. Use the Thicker/Thinner depth screw on either or both sides of the screed to adjust mat thickness as you pave. You can lock the depth screw when satisfied with mat depth. **(Figure 4-67)**
5. **C**heck the mat thickness regularly, making minor adjustments if needed.

Hydraulic Tow Point Cylinders and Grade/Slope

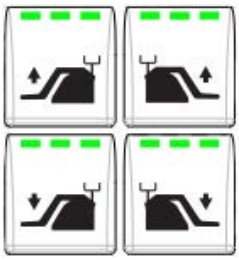
The tow point cylinders are an added convenience to the operator for adjusting the depth of the screed. The same height between the tow point and the screed pivot point above grade creates a parallel line of pull, making it possible to lay down an asphalt mat of uniformly desired thickness. This also helps the screed maintain a stable floating condition.

The 8680 paver features hydraulic tow point cylinders with linkage that allows eight inches of travel for quick and easy height adjustments, and integrates with the sonic grade and slope automation system described on ***Pages 4-53 to 4-54.*** The tow points can be operated manually or in AUTO mode.

A screed elevation gauge is located on both sides of the paver that are connected to the hydraulic tow points. **(Figure 4-68)** This gauge provides the operator with a visual reference for the tow point's position.



Figure 4-68. Screed Elevation Gauge



The left and right hydraulic tow point cylinders can be operated in manual mode from the main control panel or the screed control box. The default setting is manual. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the tow cylinders in manual mode.

Operating in Manual mode:

1. Center the tow point by referring to the screed elevation gauge. Raise or lower the screed until the rod end is flush with the "0" on the indicator decal. **(Figure 4-68)**
2. While paving, make minor adjustments to the screed using the tow point adjustment (or flight screw described in the following section).

Operating in AUTO mode:



Pressing the Grade/Slope button changes to AUTO mode. The TopCon sonic automatic grade/slope control system (included with your paver) must be connected to operate in AUTO mode. ***(Page 4-53)***

- The Tow Up and Down buttons do not function in AUTO mode unless joysticks are in neutral.
- When operating in AUTO mode, the tow points are paused when paver is not traveling forward.

4

Slope/Berm

The sloping extensions provides the means of matching curb and gutter to reduce handwork while paving. The hydraulic slope extensions provide 0 - 8 percent slope. The entire extension can be angled down to 2.5 inches.

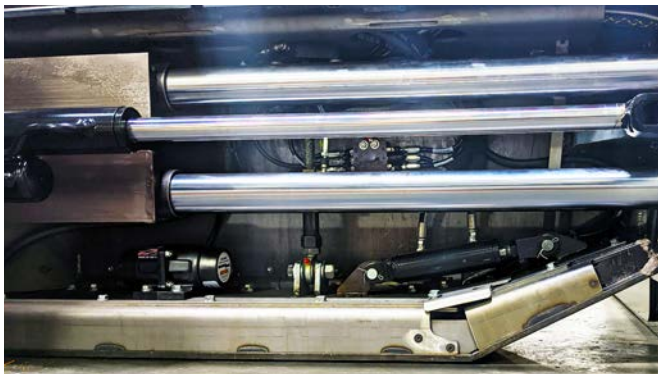


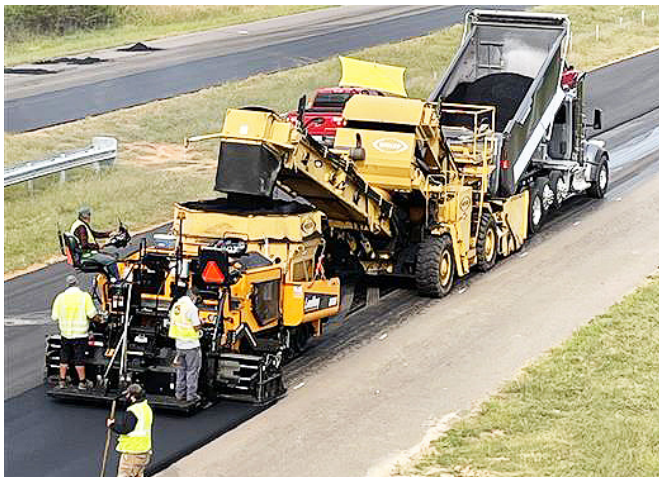
Figure 4-69. Screed Extension Berm Setting

The berm function allows you to adjust the last one foot of each extension to pave up to a six-inch roll-up curb. **(Figure 4-68)** You can raise the height to easily allow more material alongside buildings or other objects. The last foot (12 inches) of the extension can be angled up to six inches.



Use the slope/berm select button on the main control panel to choose either function, then use the slope/berm up and down buttons to increase or decrease either function.

- The default is Slope, press button for Berm. The left LED light illuminates when in slope mode and right LED light illuminates in berm mode.



STARTING TO PAVE

Inspect your paver and components to look for any issues that could affect operation or safety before work begins. Prior to startup, check all fluid levels. Start up the paver and then check all gauges.

This paver is capable of placing a mat thickness of six inches for:

- Bituminous base.
- Binder or surface course.
- Lime or Portland cement-stabilized sub-base.
- Graded aggregate materials.

Operators should be trained to operate the various machine components along with proper paving techniques before paving.

Keep these items in mind:

- Plan the project so you pave the most narrow passes first (the width of the paver), leaving the widest passes until last.
- Always pave in low gear.
- Always clean the paver after use. Remove dirt and debris from the rubber tracks and undercarriage as often as possible during the day as cleanliness directly impacts the rubber track wear rate.
- The two most important factors for laying a quality mat are all about consistency: consistent head of material in front of the screed and consistent paving speed. An increase in the head of material increases mat thickness and can translate into a wavy surface, and a decrease reduces the mat thickness and causes major flaws.

- Avoid excessive reverse operation and operating on the sides of hills/slopes as this wears tracks more quickly. Climb slopes straight up and down.
- Avoid abrupt and aggressive turns, especially at higher speeds, which puts unnecessary stress on the tracks and undercarriage. Continuous turning on the same side can also cause uneven wear on the rubber tracks.
- It is the operator's job to guide the material truck back to the hopper and signal the driver when to stop dumping asphalt into the hopper. The driver must maintain a light pressure on the brakes to prevent the truck from dumping asphalt onto the ground.
- Adjust the push rollers to the appropriate height needed for the asphalt truck. **(Figure 4-70)**
    - There are five (5) set heights on the truck push rollers. A scraper behind each one scrapes asphalt from the push rollers as you pave to help prevent asphalt build-up.
    - Remove bolts on both sides of each push roller, align with either of the five slots on the frame at desired height, and reinstall bolts.
    - Both push rollers should be at the same height to prevent damage.

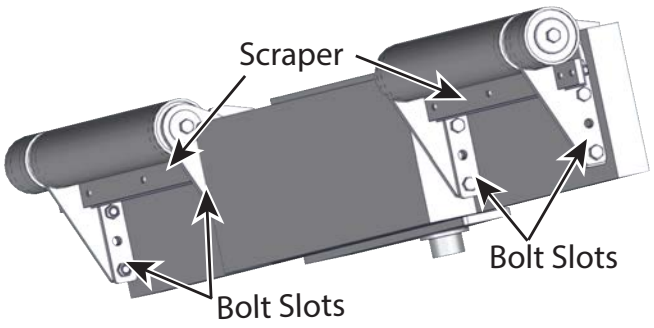
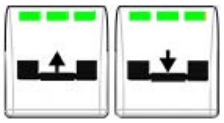


Figure 4-70. Adjustable Push Rollers



Press and hold the truck hitch IN and OUT buttons to move the truck hitch in or out as shown by arrows on the keypad symbol.

NOTE: These buttons operate the extendable [push roller if equipped with that option. ***(Page 4-67)***



Press and hold the apron UP and DOWN buttons to move the front apron up or down for additional hopper fit to the asphalt truck and reduce spillage.

- Screed operators can use the vibrator function to help with material compaction as needed while paving.



Press the button once to turn on the vibrator. Press button again to turn it off.

- It is very important that the first pass be straight as it will serve as your guideline for the following passes. Use a reference guide such as a curb, gutter, adjacent mat or string line. You can also use the screed guide at the center of the tracks on either side to further assist you in paving a straight line.
- Maintaining a precise rolling pattern is especially crucial for a quality job. Make the most number of passes while the mix is still HOT. An efficient rolling pattern that covers as much of the paved area in as short amount of time as possible is generally the best approach to achieving a uniform mat.
- If there is a line from the edge of the main screed, the screed operator needs to adjust the vertical height.
    - Lines on the inside of the main screed means the vertical is too low.

NOTE: Lines on the outside of the main screed means the vertical is too high



Automatically adjust vertical height by pressing and holding the vertical UP and DOWN buttons on the screed control box.



Basic Paving

1. Spray down the machine. ***(Pages 4-48 to 4-49)***
2. Heat screed and end gates. ***(Pages 4-55 to 5-56)***
3. Position paver at the starting point for the mat.
4. Open the Under Auger Material Cut-Offs (if equipped with this option. ***(Page 4-54)***

⚠CAUTION Never back up with cut-offs open.

5. Adjust screed as needed. ***(Pages 4-56 to 4-57)***
6. Adjust push rollers to the appropriate height needed for the asphalt truck. **(Figure 4-61)**
    - Or use extendable push roller controls if equipped with this option. ***(Page 4-67)***
7. Open hopper wings to accept asphalt from the material truck. ***(Pages 4-50 to 4-51)***

NOTE: Allow only a partial load of asphalt to enter the hopper when first starting to pave.

8. Turn the augers and conveyors on. It is recommended to operate both components in AUTO mode (default). ***(Pages 4-51 to 4-54)***
    - Ensure the sonic auger sensor is connected for AUTO operation.
    - When the material starts moving into the sonic range, the sonic auger will automatically regulate auger operation.

NOTE: Bypass auger operation when paving a basic eight-foot pull as auguring is not necessary at this width.

9. When the material begins to discharge from under the screed, press the Screed Down button to float the screed.
10. Begin paving, moving slowly at first, adjusting the screed if needed.
    - Use the hydraulic tow points, grade/slope functions, and/or depth screws as needed while paving. ***(Page 4-59)***
    - The procedures for setting the crown and making AOA (Angle of Attack) adjustments for asphalt mat texture are described on the following pages.
11. Make a paving pass until the asphalt level in the hopper is low.
12. Return paver back to starting position to begin the next pull.

Setting the Crown

The screed can also be set to control the pavement valley, or crown. A pavement crown is the elevation of the middle in relation to the edges, and insures proper drainage of the pavement.

NOTE: Maximum crown is two inches.

To set the crown:

1. Loosen the lock-down bolts in slotted bars before adjusting crown and valley mechanism. **(Figure 4-71)**



Figure 4-71. Loosen Lock-Down Bolts

2. Use the crown handle adjuster for setting positive or negative crown:
   - For increased positive crown push down on adjuster.
   - For increased negative crown pull up on adjuster.
   - To put leading edge crown in the screed, you must make the adjustment with the crown handle. Once leading edge is set where you want, remove the four locking bolts, then set the trailing edge flat and reinstall the locking bolts.
3. Use the gauge at the center of the screed above the screed platform as a visual reference.
   - If the needle is above the zero, you will have positive crown.
   - If the needle is below the zero, you will have negative crown.
4. To get exact crown or valley, measure the distance between a flat level surface to the center bottom portion of screed. Make adjustments with crown and valley control.
   - Positive crown is when the middle of the mat is raised to permit water to drain to each side. **(Figure 4-72)**
   - Negative crown is the lower in the center of the screed plate, often used in an alley where drainage is necessary.

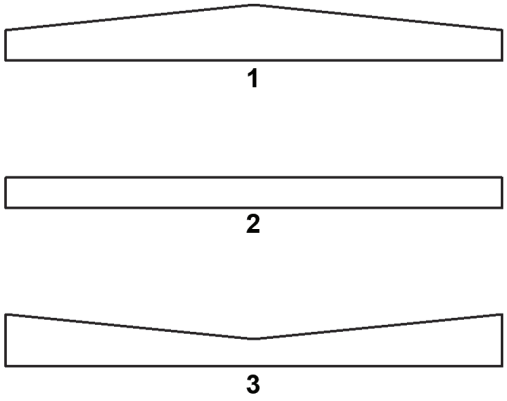


Figure 4-72. Crown Settings

1 - Positive (+)

2 - Zero (0)

3 - Negative (-)

NOTE: Crown may be placed in the leading edge and/or the trailing edge of the screed plate. Crown in the leading edge aids material flow under the screed plate only. Trailing edge crown puts a crown in the mat.

Example: The trailing edge crown is 0 and the leading edge crown is 1/8 inch. There will not be any crown in the mat, but the material flow under the screed plate will be improved.

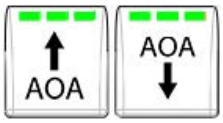
Mat Texture Adjustment

The Angle-of-Attack (AOA) is the angle between the bottom of the screed plate and the grade surface. Adjusting the AOA produces a smooth or coarse mat texture.

- The screed extension should be hot before making any adjustments.
- Spray the bottom of the screed with cleaning solvent BEFORE heating the screed.

⚠CAUTION NEVER spray cleaning solvent or release agent on or near a hot screed heating or near open flame. Cleaning solvent and release agents are flammable and could cause serious personal injury.

NOTE: Do not allow release agent chemicals to run onto the ground.



The screed operator can make AOA adjustments using the AOA Up and Down buttons on the screed control box while paving.

1. Spray the heat screed extension with release agent or solvent if needed.
2. Press and hold the AOA Up and Down buttons to increase or decrease the AOA:
   - Pressing the AOA Up button increases the pressure on the back of the extension, which gives a smoother, slicker finish.
   - Pressing the AOA Down button decreases the pressure on the back of the extension, resulting in a coarser finish.

NOTE: Too much pressure on the back of the screed extension will take the weight off the screed wear plate and cause poor material compaction, resulting in a poor finish.

UNLOADING AND LOADING

Ensure the transport trailer has the capacity to carry the paver's weight. Park the trailer in a clear, level area when loading or unloading the machine.

⚠CAUTION ALWAYS position an assistant to be a ground guide when loading and unloading the paver.

⚠CAUTION Work slowly and carefully to avoid accidents. Keep the loading area clear.

⚠CAUTION ALWAYS raise the augers when loading the paver onto transport vehicles.

⚠WARNING ALWAYS back the paver up loading ramp with the screed end first. ALWAYS be alert to uneven surfaces and the potential reactions when transferring the paver from surface to surface. Serious injury or even death can result.

⚠WARNING Use extra caution steering when loading and unloading the paver as slight steering changes on ramps and trailers can cause the machine to fall off the ramp or transport trailer. Serious injury or even death can result.

⚠CAUTION To prevent damage to the undercarriage and throughout the paver, reduce traveling speed before the paver tracks come into contact with loading ramps or any abrupt change in the surface. The track drive sprocket or other components can be damaged by excessive shock.

⚠CAUTION DO NOT allow the screed to strike the ramp as this can damage bearings on the thickness control screws or welds on the leveling arms. A longer ramp to reduce the loading angle may be needed.

NOTICE Never move the paver in reverse while the cut-offs are open.

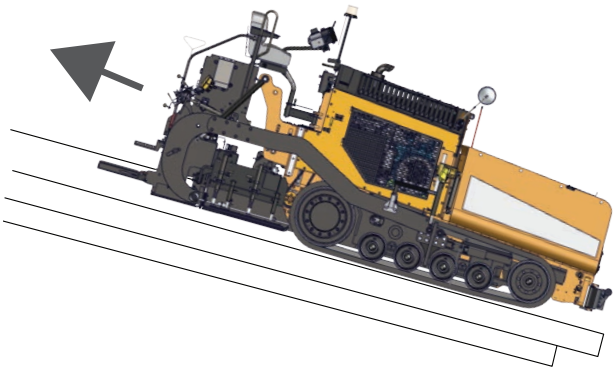


Figure 4-73. Correct Unloading and Loading Position

Unloading

1. Remove tie-down equipment.
2. Start and warm up engine.
3. Set throttle to Pave Low (1800) RPMs.
4. Set steering control lever so paver moves very slowly.
5. BEFORE moving the machine, ensure:
   - The screed position is UP.
   - Auger extensions are removed.
   - The screed extensions are IN.
   - The Under Auger Material Cut-Offs below the augers are CLOSED.
6. Move paver forward down the ramp.

Loading

⚠CAUTION **The paver must be loaded screed-end first to prevent damage.**

1. Back up paver to base of the ramp, aligning tracks with the ramp. **(Figure 4-73)**
2. BEFORE moving machine up the ramp, ensure:
   - The screed position is UP.
   - Auger extensions are removed.
   - The screed extensions are IN.
   - The Under Auger Material Cut-Offs are CLOSED.
3. Load the paver screed-end first. Set throttle to Pave Low (1800) RPMs.
4. Set steering control lever so paver moves very slowly onto the ramp.
5. Using the steering control joysticks Or steering knob if equipped), SLOWLY guide paver up the ramp.
6. Position paver in center of trailer.
7. Lower the screed.
8. Shut down the engine.
9. Attach tie-down chains to the front weight block and D-Rings in front of the torque hubs. .
10. Place chocks at tracks.
11. Ensure that all chains are tightly secured before transporting the paver.

TOWING

⚠WARNING **Ensure the paver is on level ground and blocked to prevent machine from moving while preparing for towing. Be especially careful after disconnecting the torque hub as the paver could roll.**

While it is strongly recommended that the paver be transported rather than towed, there are situations where you may need to tow the paver off the road until a transport vehicle arrives. **DO NOT tow paver long distances.**

You must manually disconnect the brakes using a hydraulic pump to prepare the machine for towing. The brake release mechanisms are located inside the right engine access door. **(Figure 4-74)**

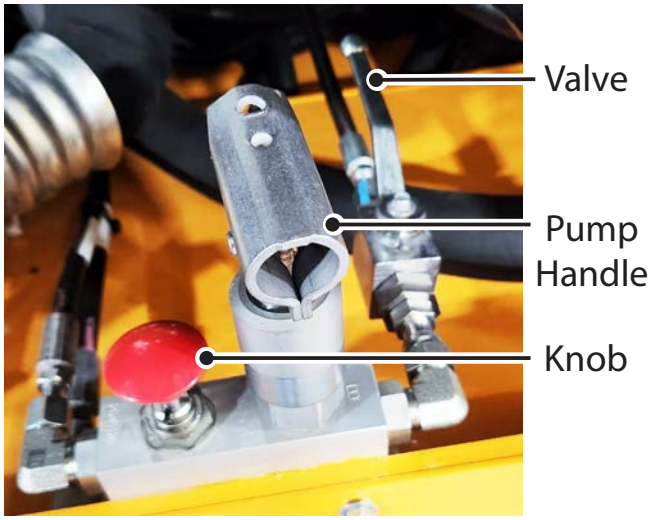


Figure 4-74. Brake Release Pump Valve

To release the brake for towing:

1. Push the red knob into the lower position.
2. Pull the valve handle 1/4 turn counterclockwise.
3. Using the pump handle, pump the brake release until tight. Brakes are now released for towing.
4. When the paver has been towed to its destination, open the valve by turning 1/4 turn clockwise to re-engage brakes.

OPTIONAL COMPONENTS

LB7 Screed

The monumental LB7 screed is packed with easy-to-use, highway class features for screed operators. **(Figure 4-75)** This versatile screed with front mounted extensions can pave up to 15.6 feet. A 34kW generator provides power to the S-curve heating elements on the screed extensions. With four independently user-adjustable heat zones, the operator is given precise and automatic heat to the zone as needed.

The extensions are equipped with electronic, on-demand vertical height and angle-of-attack (AOA) adjustments with the press of a button. This screed is equipped with spring-loaded end gates, giving the operator the downward pressure needed when matching joints or against curbs and gutters.

Additional options for the LB7 screed include:

- 20-foot Paving Kit (also available for the LB5 Screed)
- 18x16-inch Berm Extensions
- 12x6-inch Berm Extensions



Figure 4-75. Optional LB7 Screed

Umbrella Option

The umbrella kit option provides shelter from sun and other weather conditions for the operator. The umbrella kit assembly includes the mounting bracket and umbrella frame assembly that can be mounted on either side of the screed handle.

Screed Extension Vibration

Screed vibration helps compact material. If equipped with this option, use the Vibrators button on the main control panel to turn vibration on and off.

Screed Central-Point Lube System

The screed central-point lube system combines all screed lubrication points into one. When you pump grease into the central point, valves distribute the grease evenly to all the screed grease points.

Single Point Conveyor Lube Kit

The Conveyor Lube kit combines the 10 conveyor bearing grease points and the two front wheel spindle bearings. This kit combines these 12 individual grease points into one lubrication point.

Steel Poly Tracks

Steel poly tracks are available in lieu of the rubber tracks. Choose this option at purchase of the 8680 paver.

Steering Knob

The steering knob option is used for steering and replaces the right joystick on the control panel. **(Figure 4-76)**



Figure 4-76. Steering Knob Option

Steering Wheel

A steering wheel option is also available in lieu of joystick steering. **(Figure 4-77)** The Run/Stop switch on the steering panel controls running and stopping the machine, and applies parking brake when set to the STOP position. The joystick controls forward and reverse travel along with speed.



Figure 4-77. Steering Wheel Option

- To drive the paver forward, push joystick forward to reach the desired speed.
- To move in reverse, pull joystick(s) and move backwards to reach desired speed. A back-up alarm will automatically turn on when in reverse.
- To slow the paver, move joysticks closer to the neutral position.
- Return joystick to the neutral position to stop paver.

NOTE: If the joystick steering function is not being used, the Run/Stop switch must be in the center OFF position.

NOTE: The engine will not start unless the E-STOP button is disengaged and the joystick must be in the NEUTRAL (center) position.

20-Foot Paving Kit

The 20-foot paving kit includes all components for enabling your paver to lay asphalt mats up to 20-feet wide. This option is available for both the LB5 and LB7 screeds, and must be selected when ordering the 8680 Commercial Paver.

Berm Extensions

Berm extensions extend the screed's functionality to forming a sloped asphalt berm or shoulder, which can be crucial for drainage and meeting project specifications. Available only with the LB7 screed option, the berm extensions are available at 18" x 6" or 12" x 6" dimensions.

High-Lift End Gate

The high-lift end gate option is a specialized feature that helps create a defined, high-quality pavement edge, even when the screed is extended to its maximum width or encountering obstructions. The high-lift end gate can be extended to ride on top of the curb, ensuring a smooth and consistent transition between the paved surface and the curb. They are especially useful when working on projects with different paving widths, winding roads, or situations where the end gate needs to be extended.

Extendable Push Roller

The extendable, oscillating push roller option offers even more flexibility for maintaining contact with an asphalt material truck. **(Figure 4-78)** Extending the rollers various lengths as needed allows better contact for material trucks of all sizes, decreasing the load on the paver so it's easier to push the truck. Use the same procedures in the previous section for operation.

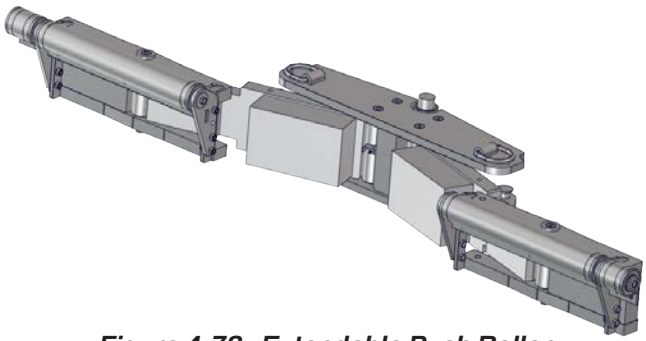


Figure 4-78. Extendable Push Roller

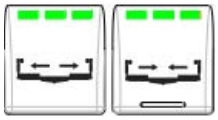
Truck Hitch

The truck hitch is an optional attachment that prevents excessive and uneven braking from the asphalt material truck. **(Figure 4-79)**



Figure 4-79. Truck Hitch

To engage the hitch with the rear wheels of the asphalt truck:



1. Use the Truck Hitch button on the main control panel to move truck hitch into the desired position.
2. Extend the arm extensions of the truck hitch by pushing and holding the Truck Hitch button in the OUT position.
3. The paving material truck then SLOWLY backs up to the paver until the hitch makes contact with the rear tires of the truck.
4. Retract the arm extension by pushing and holding the Truck Hitch button in the IN position until both guide rollers are fully locked onto the truck wheel rims.

NOTE: It may be necessary to adjust the roller guides to the inside of the wheel rims.

Topcon® Sonic and Dual Grade Control

Topcon® sonic grade control, dual grade control and slope options offer paving professionals screed control for improving smoothness along with assuring accurate slope and material thickness. Sonic trackers continuously measure the elevation over the surface, automatically averaging and adjusting the tow point cylinder on the paver. These controllers send signals to the screed to precisely maintain mat thickness and slope as you pave. Most states require this technology for state road paving jobs to ensure smoothness and continuity on roadways.

Choose from one of three options: Sonic Grade Control (one side); Dual Grade Control or Dual Grade Control and Slope. The dual grade controls automatically maintain a set depth on both ends of the screed. The Dual Grade and Slope controls maintain a set depth on one end of the screed and a set slope on the other end.

The paver must be pre-wired for TopCon technology. If equipped with this option, be sure to unplug and store the system after use. Always recap the cables to protect the connectors when not in use. **(Figure 4-80)**



Figure 4-80. TopCon Sonic Grade Control Box

To initiate the TopCon system:

1. Start the engine and idle a few minutes to warm hydraulic oil throughout the machine.
2. Connect the plug into the connector the main control panel column. (The plug will only fit into its correct connector.)
3. Push the power button to turn the system ON.
4. Turn the Mode Select Knob clockwise (or counterclockwise depending on last setting used) until the number 14 appears in the Mode Select display window.
5. Press and hold the Set button for three seconds and release.
6. Press the Set button again.
7. Turn the Mode Select Knob until EXT (external) appears in the Mode Select display window.
8. Press Set button again, then press Menu button.
9. Turn the Mode Select Knob until the number 50 appears in the Mode Select display window.
10. Repeat Steps 5 and 6 to set.
11. Turn the Mode Select Knob until the number 16 appears in the Mode Select display window. If the tow arm cylinder does not begin to move, continue turning the knob until it does. **DO NOT raise or lower the screed.**

NOTE: This is the valve offset setting, which varies from paver to paver. Make a note of this number on the Notes page provided in this manual for future reference.

12. Repeat Steps 5 and 6 to set.
13. Turn the Mode Select Knob until the number 14 appears in the Mode Select display window. If the tow arm cylinder does not begin to move, continue turning the knob until it does.

NOTE: This is the elevation setting, which varies from paver to paver. Make a note of this number on the Notes page provided in this manual for future reference.

NOTICE **Setting this value too high will result in overshooting the grade.**

14. Repeat Steps 5 and 6 to set.
15. Turn the Mode Select Knob until the number 15 appears in the Mode Select display window.
16. Press the Set button. You are now ready to pave.

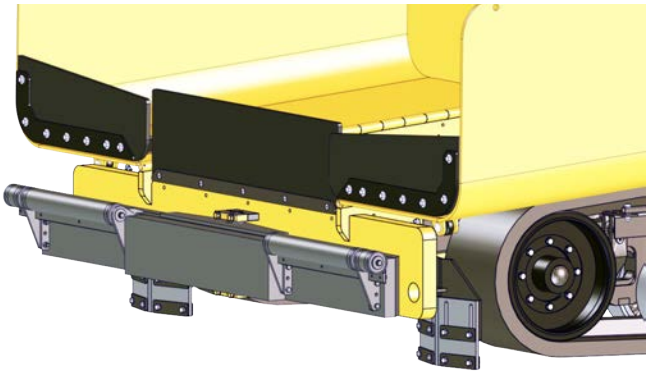


Figure 4-81. Track Plows

Track Plows

Track plows mounted on the frame in front of the tracks shield tracks from asphalt, dirt and debris. **(Figure 4-81)** See your authorized LeeBoy dealer for purchase.

Power Crown Option

The Power Crown option adjusts the positive or negative crown as needed. The power crown switch is located on the junction box. **(Figure 4-82)** (It is included with the LB7 screed option.)

- Push the switch UP for valley.
- Push the switch DOWN for crown.



Power Crown Switch

Figure 4-82. Power Crown Switch

Slope Meter

The slope meter is attached to the screed (beside the crown adjuster lever). The slope meter is used for slope and crown road construction, and continuously indicates the slope of the ground, or "percent of grade." **(Figure 4-83)**



Figure 4-83. Slope Meter

The slope meter is calibrated to read from .00 to .12 (0 - 12 percent) and works similar to a level tool. Just like a level tool, it contains a ball bearing suspended in oil that gravity always keeps at the lowest point of the meter, indicating the slope as you pave. Fluid in the vial is heavy and helps keep the indicator ball steady even with vibration, yet it still reacts quick enough to keep you accurate. While the slope meter is sturdy and functions regardless of weather conditions, take care not to break the glass tube. Keep the slope meter clean for optimum viewing at a glance.

If you selected the slope meter option when your ordered your paver, it is already mounted onto the screed at our manufacturing facility. Otherwise, you will need to mount the slope meter onto the screed.

NOTE: The slope meter is a precision instrument and must be installed correctly for accurate readings. Install the slope meter onto the metal block (if equipped), or a solid, straight, flat area on the main screed frame that will not twist or strain during operation. It cannot be emphasized enough that the paver needs to be on level ground for correct installation and operation.

To install the slope meter:

1. Park the paver on a level surface. Use a level tool to determine that the paver is level before installing.
2. The slope meter comes bolted to an aluminum mounting plate (two bolts). Remove the bolts and the slope meter from its mounting plate.
   - If your paver is equipped with a metal block welded onto the screed frame, discard the mounting plate as you will be attaching the slope meter directly to the metal block surface.

NOTE: Pay attention to the raised rims around the bolt holes. The slope meter must be remounted on the same side to prevent damage during installation.

3. Align top edge with the frame edge. Use a level to ensure it is perfectly level.
   - If using the aluminum plate, align it onto the mounting surface carefully, with the top edge of the plate aligned with the frame edge.
   - If installing onto the metal block, use a level tool to ensure
4. Reinstall the two bolts securely, ensuring the raised rims are in their original position.

NOTE: If using the aluminum plate, the lock washers provided MUST be used or the plate can shake loose.



Figure 4-84. Balloon Light

Balloon Light

The bslloon light is mounted on the operator platform to illuminate the work area when paving at night. **(Figure 4-84)** This LED light provides strong illumination in all directions around the paver. The light can also be adjusted for height, then easily retracted for storage. See your authorized dealer for purchase

Magnetic Cup Holder

Magnetic cup holders can be purchased for use with your machine (and the rest of your fleet). **(Figure 4-85)**

See your authorized dealer for details. Ask for **Part Number 1020521.**



Figure 4-85. Magnetic Cup Holder

NOTES

Section 5

MAINTENANCE

Page

Maintenance 5-1

Maintenance Schedule 5-5

Accessing Components 5-6

Raise and Lower Engine Hood 5-6

Routine Maintenance 5-7

Prepare Paver for Maintenance 5-7

Every 10 Hours or Daily Routine Maintenance 5-7

Every 50 Hours or Weekly Routine Maintenance 5-8

Every 100 Hours or Monthly Maintenance 5-8

Every 250 Hours or Quarterly Routine Maintenance 5-8

Every 500 Hours or Semi-Annual Routine Maintenance 5-8

Every 1000 Hours or Annual Routine Maintenance 5-8

Every 1500 Hours 5-8

Every 3000 Hours 5-8

Lubrication 5-9

Lubrication Points Chart 5-9

Torque Hub Lubrication 5-13

Maintenance Adjustments 5-14

Conveyor Flight Chains 5-14

Conveyor Drive Chains 5-15

Auger Drive Chain 5-15

Engine Maintenance 5-16

Check Engine Lubrication Oil 5-17

Change Engine Oil and Filter 5-17

Change Air Filter 5-18

Engine Drive Belt 5-19

Change Oil Separator Element 5-19

Selective Catalytic Reduction (SCR) 5-20

DEF (Diesel Exhaust Fluid) 5-20
DEF Storage 5-21
DPF Service 5-21
Cooling System 5-22
Replace Radiator Hoses 5-22
Fuel System 5-23
Fuel Tank 5-23
Change Fuel Filter 5-23
Fuel Water Separator 5-24
Hydraulic System 5-24
Check Hydraulic Oil 5-24
Change Hydraulic Oil and Filters 5-25
Electrical System. 5-26
Battery Servicing. 5-26
Alternator 5-27
Adjust Generator Belt 5-27
Fuse Panel. 5-28
Electric Heat Control Box. 5-28
Heating Elements 5-30
Element Resistance Testing 5-30
Generator Speed Check 5-31
Generator Voltage Testing 5-31
Dielectric Grease 5-32
Drive System. 5-33
High-Speed Rubber Tracks 5-34
Track Tension Cylinder 5-35
Oscillating Bogies 5-35
Front Idler 5-36
Two-Speed Hydraulic Motors 5-36
Planetary Torque Hub 5-37
Replacement Procedures 5-38
Hydraulic Pump/Motor Start-Up 5-39
Rear Conveyor Shaft, Bearings and Sprockets 5-40
Conveyor Drive Motors. 5-41
Front Hopper Wing Cylinders 5-42
Rear Hopper Wing Cylinders 5-42

Auger Motors 5-43
Auger End Mounts 5-44
Auger Inner Bearings 5-44
Main Screed Wear Plate 5-45
Screed Extension Heat Box 5-47
Winterizing for Storage 5-49
Troubleshooting 5-50
Hydraulic Pressure 5-50
Track Tension Pressure Relief 5-50
Spray Down Pump 5-51
Conveyor Chain Tension Slack Adjustment 5-52
Troubleshooting Chart 5-53
Display Diagnostic Trouble Code (DTC) Chart 5-57

NOTES

MAINTENANCE SCHEDULE

Before performing any maintenance procedures on the LeeBoy Model 8680 Commercial Paver, review the safety information in **Section 1**. Refer to the Engine Operator's Manual accompanying your paver for more detailed engine service information.

⚠WARNING Always use the appropriate and correct sized tools for the task at hand to prevent damage or possible injury.

⚠CAUTION When performing any routine maintenance, always include the previous routine maintenance hours in the higher hourly schedule.

Table 5-1. Periodic Maintenance Schedule

SYSTEM	ITEM	Every 10 Hours Daily	Every 50 Hours Weekly	Every 100 Hours Monthly	Every 250 Hours Quarterly	Every 500 Hours Semi-Annually	Every 1000 Hours Annually
Paver	Lubricate paver.	X					
Engine	Check oil level.	X					
	Check battery for corrosion and clean if needed.	X					
	Check coolant level.	X					
	Drain and replace coolant.					X	
	Drain fuel water separator if needed.	X					
	Clean fuel water separator.					X	
	Inspect/clean air filter.				X		
	Change air filters.						X
	Replace fuel filter.				X		
	Change oil and oil filter. **(Initial 50 Hours)**	X			X		
	Inspect/adjust engine belt.				X		
	Replace engine belt.					X	
	Check DEF tank fluid level. Fill if needed.	X					
	Check/adjust generator belt.			X			
Hydraulic	Check hydraulic oil level.	X					
	Change hydraulic oil and filter. **(Initial 50 Hours)**	X			X		X
	Check torque hub oil. Refill if needed.		X				
	Change torque hub oil.					X	
	Drain hydraulic tank. Replace strainer.						X
Mechanical	Check/adjust conveyor drive chains.			X			
	Check/adjust conveyor flight chains.			X			
	Check/adjust auger chains.			X			
	Check/adjust screed extension top guide.			X			
	Contact dealer to check valve clearance.						X
Every 1500 Hours	Replace oil separator element. Contact your authorized dealer to check engine injector tip, PCV (Positive Crankcase Ventilation) valve, and EGR cooler.						
Every 3000 Hours	Contact your authorized dealer to check EGR system and turbo-charger.						

5

ACCESSING COMPONENTS

Raise and Lower Engine Hood

Accessing the radiator and other engine components is simple and convenient on the 8680 commercial paver. The engine hood pivots forward for convenient access to the radiator and other engine service points. **(Figure 5-1)** Access doors on each side of the paver also provide easy access for servicing various components.

To raise the hood:

1. Push the center of the hood latches (5) across the **(Figure 5-1)** to unlatch. (One side of the latch mechanism pops up.)

⚠CAUTION Ensure all hood latches (5) are unlatched before raising the hood.

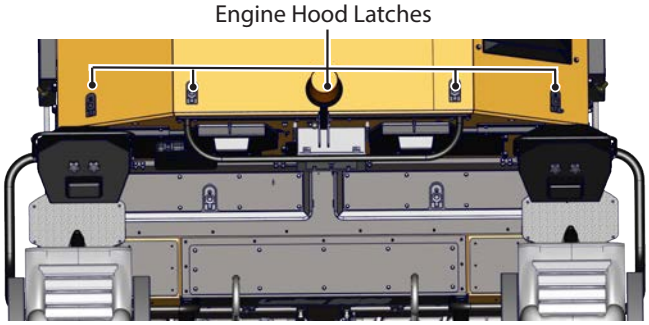


Figure 5-1. Engine Hood Latches

2. Turn the ignition switch to the RUN position. **(Figure 5-2)**
3. Use the Hood Lift switch on the junction box to raise and lower the hood:
   - Push the Hood Lift switch UP and hold to raise the hood.
   - Push the Hood Lift switch DOWN to close the hood.
4. After lowering the hood completely, push the latches back into place to secure the hood.

⚠CAUTION Ensure all latches are securely latched before operating.

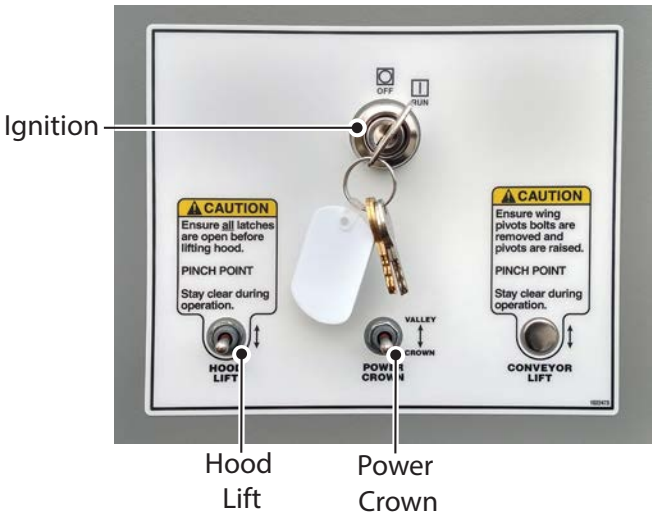


Figure 5-2. Junction Box Controls

NOTE: The conveyor lift switch is nonfunctional on this model.

ROUTINE MAINTENANCE

Following the maintenance schedules and procedures will maintain the paver in top operating condition and provide years of trouble-free operation. Refer to the Engine Operator's Manual accompanying your paver for more detailed engine service information.

NOTICE **Changing oil and cleaning the paver should only be done in a designated area where the oil and chemicals can be contained. Discard these by-products in accordance with environmental regulations.**

When performing any routine maintenance, always include the previous routine maintenance hours in the higher hourly schedule.

Prepare Paver for Maintenance

⚠WARNING **If any maintenance must be performed while the engine is running, DO NOT leave paver unattended.**

1. Park the paver on a flat surface.
2. Lower all attachments to ground level.
3. Place joysticks in neutral.
4. Run engine at 1/2 speed (RPMs) without load for three to five minutes, then reduce to slow idle.
5. Place ignition switch in the OFF position.

NOTICE **If asphalt material is allowed to remain on the paver overnight, possible damage and increased maintenance costs can result upon start-up the next day. Spray down the paver daily (or more often if needed).**

Every 10 Hours or Daily Routine Maintenance

1. Clean off paving material and spray cleaning solvent or release agent on the wear plate, extensions, and any other area that has been in contact with the paving material. Cleaning should be performed while the paver is hot.

⚠WARNING **NEVER spray cleaning solvent on hot surfaces as serious injury can result.**

2. Lubricate the required parts. ***(Pages 5-9 to 5-13)***
3. Check battery for corrosion and clean if needed. Check cables and connections. ***(Page 5-26)***
4. Check coolant level. Fill if needed. ***(Page 5-22)***

⚠WARNING **Allow the radiator to cool off completely (30 minutes after engine operation) before checking coolant level. NEVER remove the radiator cap when hot to prevent serious burns.**

5. Check DEF tank fluid level. Fill if needed. ***(Page 5-20)***
6. Check for loose, worn or damaged hoses, tubes clamps. Tighten, repair or replace promptly.
7. Check fuel level. Keep the fuel tank full to prevent condensation. Always fill at the end of the day. ***(Page 5-23)***

⚠WARNING **ALWAYS stop the engine when refilling fuel and be careful not to spill to prevent a fire. DO NOT smoke while refueling. Only add fuel in well-ventilated area or outdoors. DO NOT overfill.**

NOTICE **Use only ultra low sulfur diesel fuel.**

8. Check engine oil level. ***(Page 5-17)***
9. Check hydraulic oil level. ***(Page 5-24)***
10. Drain fuel water separator if needed. ***(Page 5-24)***
11. Perform any other engine preventive maintenance described in the Engine Operator's Manual.

NOTE: In cold weather, keep conveyor flight chains properly oiled with cleaning solvent or release agent. This will prevent conveyor bars from sticking. Neglect could result in conveyor bar damage or drive chain failure.

Every 50 Hours or Weekly Routine Maintenance

1. The FIRST 50 HOURS, change engine oil and filters, then every 250 hours thereafter. ***(Page 5-17)***
1. The FIRST 50 HOURS, change hydraulic oil and filters, then every 250 hours thereafter. ***(Page 5-25)***
2. Check torque hub oil level. Add more oil if needed. ***(Page 5-13)***
3. Lubricate the flight screws, extension cylinders, end gate adjusters and crown/valley mechanism. Grease nuts on extension screws. ***(Pages 5-10 to 5-12)***
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 100 Hours or Monthly Maintenance

1. Check generator belt. Adjust if needed. ***(Page 5-27)***
2. Check and adjust conveyor drive chains. ***(Page 5-15)***
3. Check and adjust conveyor flight chains. ***(Page 5-14)***
4. Check auger drive chains, lubricate and adjust.
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 250 Hours or Quarterly Routine Maintenance

1. Inspect/clean air filter element. ***(Page 5-18)***
2. Inspect/adjust engine belt. ***(Page 5-19)***
3. Change engine oil and filter. ***(Page 5-17)***
4. Change hydraulic oil and filter. ***(Page 5-25)***
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 500 Hours or Semi-Annual Routine Maintenance

1. Drain radiator and replace coolant. ***(Page 5-22)***
2. Change torque hub oil. ***(Page 5-13)***
3. Replace engine belt. ***(Page 5-18)***
4. Replace air filter. ***(Page 5-19)***
5. Clean fuel water separator. ***(Page 5-24)***
6. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 1000 Hours or Annual Routine Maintenance

1. Drain hydraulic tank and refill with fresh hydraulic oil. Change strainer. ***(Page 5-25)***
2. Replace air filter. ***(Page 5-18)***
3. Contact your dealer to check valve clearance.
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 1500 Hours

1. Contact your authorized dealer to check and service (if needed), the engine injector tip, PCV (Positive Crankcase Ventilation) valve, and EGR cooler.
2. Replace oil separator element. ***(Page 5-19)***
3. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 3000 Hours

1. Contact your authorized dealer to check and service (if needed) the EGR system and turbo-charger.
2. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

LUBRICATION

Proper lubrication is extremely important and necessary to maintain the paver and its components at top efficiency. Lubrication reduces wear on moving parts, seals and protects components, absorbs shock, minimizes corrosion and reduces operating temperatures.

NOTICE **Lubrication service should only be done in a designated area where the oil and chemicals are contained. These by-products should be cleaned up and discarded in accordance with environmental regulations.**

Lubrication Points Chart

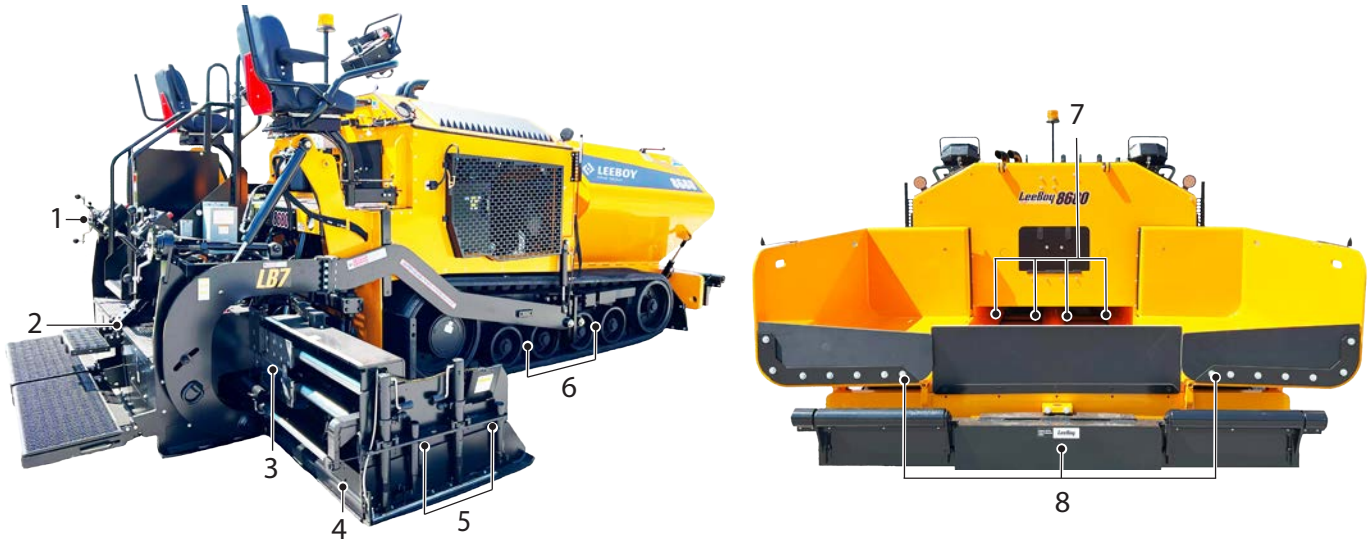


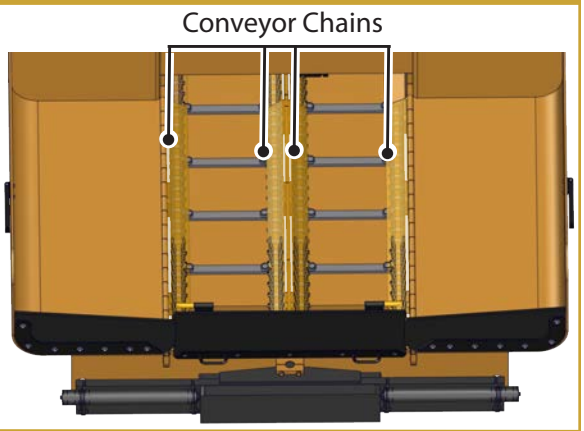
Table 5-1. Lubrication Points Chart

ITEM NO.	TYPE LUBE	DESCRIPTION AND LOCATION	INTERVAL
	A	Grease With Shell Avania EP Grease 2 or Equivalent	
	B	Spray with Release or Chain Lube Agent	
	C	Spray with Copper-Based Anti-Seize Agent	
1	A	Flight screw (both sides). Grease first in locked position, unlock, and turn 180° to grease. Also lube U-Joints.	Weekly
2	A	Crown and Valley.	Weekly
3	C	Screed extension slide plates (both sides).	Weekly
4	A	All screws and bearings on screed extensions (both sides).	Weekly
5	A	End gate adjusters (both sides).	Weekly
6	A	Track pivots (both sides).	Weekly
7	B	Conveyor chains (one on each side of each conveyor).	Daily
8	B	Spray all parts of paver that come into contact with asphalt.	Daily

Daily Lubrication

Spray these components with release agent daily:

- Conveyor chains (one on each side of each conveyor).
- Any part of the paver that comes into contact with asphalt.



Spray both screed extension slides with copper-based anti-seize agent ONLY (LB5 and LB7 screeds).

DO NOT spray release or chain lube agent onto the screed extension rods and slides.

NOTE: Keep the chrome rods clean, but DO NOT lubricate.

